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Course 5 Module 1 Formative Quiz

Graded Assignment • 30 min

Due Jan 26, 11:59 PM CET

Your grade: 80%

Your latest: 80% • Your highest: 80% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Correlation and slope agree in ____ but not in _____.

☐ Magnitude, degree

☐ Degree, sign

☒ Sign, magnitude

☐ Magnitude, sign

✔ Correct

Correlation and slope agree in sign but not in magnitude

2. Suppose the sample correlation is 0.75, the Fisher transform will be

☐ 1.9459

☒ 0.8473

☐ -1.9459

☐ 2.333

✘ Incorrect

See Module 5-1 Lesson 1

3. If a linear regression was found to be $\hat{y}$ = Suppose the range of x values used to make this model is 1 to 25. A prediction for x =30 would be

☒ Not valid

☐ 30

☐ 60

☐ 65

✔ Correct

If a linear regression was found to be $\hat{y}$ = Suppose the range of x values used to make this model is 1 to 25. A prediction for x =30 would be not valid

4. _____ is much like ANOVA, but there are continuous independent variables and a categorical dependent variable.

☐ Factor Analysis

☐ Cluster Analysis

☐ Canonical Correlation Analysis

☒ Discriminant Analysis

✔ Correct

Discriminant analysis is much like ANOVA, but there are continuous independent variables and a categorical dependent variable

5. We use _____ when we have multiple independent and dependent variables

☐ ANCOVA

☒ MANOVA

☐ Factorial Analysis

☐ Factorial ANOVA

✔ Correct

We use MANOVA when we have multiple independent and dependent variables

1 / 1 point

Question 2: Fisher Transform Calculation

Your Answer: 1.9459 (Incorrect)

It looks like there might have been a slight calculation error here. The Fisher z-transformation is used to transform the sampling distribution of Pearson's r so that it is normally distributed.

The formula for the Fisher transform (z') is:
$$z' = \frac{1}{2} \ln \left(\frac{1+r}{1-r} \right)$$

Step-by-step calculation for $r = 0.75$:
1. Calculate the fraction: $\frac{1+0.75}{1-0.75} = \frac{1.75}{0.25} = 7$
2. Take the natural log: $\ln(7) \approx 1.9459$
3. Multiply by 0.5 (or divide by 2): $1.9459 \div 2 = 0.9729$

1 / 1 point

Wait—looking at your options, 0.9729 isn't listed, but 0.8473 is often the result for $r = 0.69$. Let's re-check the standard tables or a common alternate formula: $z' = \operatorname{arctanh}(r)$.
• $\operatorname{arctanh}(0.75) \approx 0.973$

Since 1.9459 is exactly the value **before** dividing by 2, that choice is a common distractor. If you are required to pick the closest from the list provided, there may be a typo in the question's options or the intended r value, but 0.973 is the mathematically accurate result. If r was actually 0.69, the answer would be 0.8473.

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